# Board A — UHF front end + calibration noise source

Power: 12 V in, 5.6 V buck, low-noise 5 V and 3.3 V

File: power.kicad_sch

Calibration noise source and 6-way splitter

File: noise_source.kicad_sch

RF channels 0–2

File: channels_0.kicad_sch

RF channels 3–5

File: channels_1.kicad_sch

PWR_FLAG

+12V

PWR_FLAG

+5V_LNA

PWR_FLAG

GND

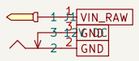
Sheet: /
File: board_a.kicad_sch

**Title: Board A — UHF front end + calibration noise source**

| | | |
|---|---|---|
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.3 | | Id: 1/5 |

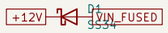
# Power: 12 V in, 5.6 V buck, low-noise 5 V and 3.3 V

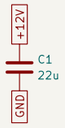
CAL_EN (from the digitizer/FPGA, 1.8–3.3 V logic): high = noise source on and all switches on RF1 (cal).

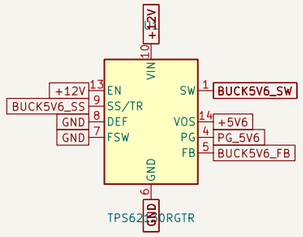








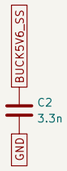


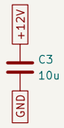


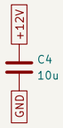


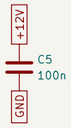


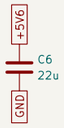


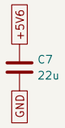


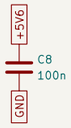



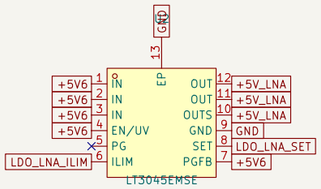



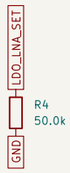





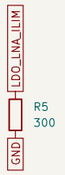


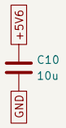








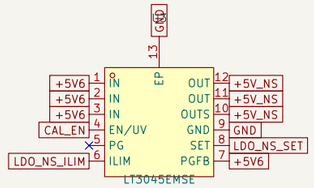


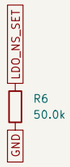


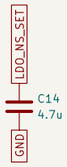


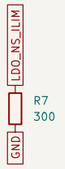


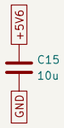


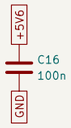


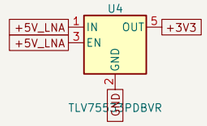



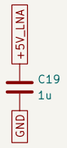



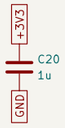


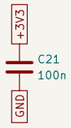




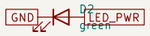



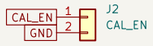







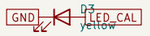





















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**Passive radar receiver**

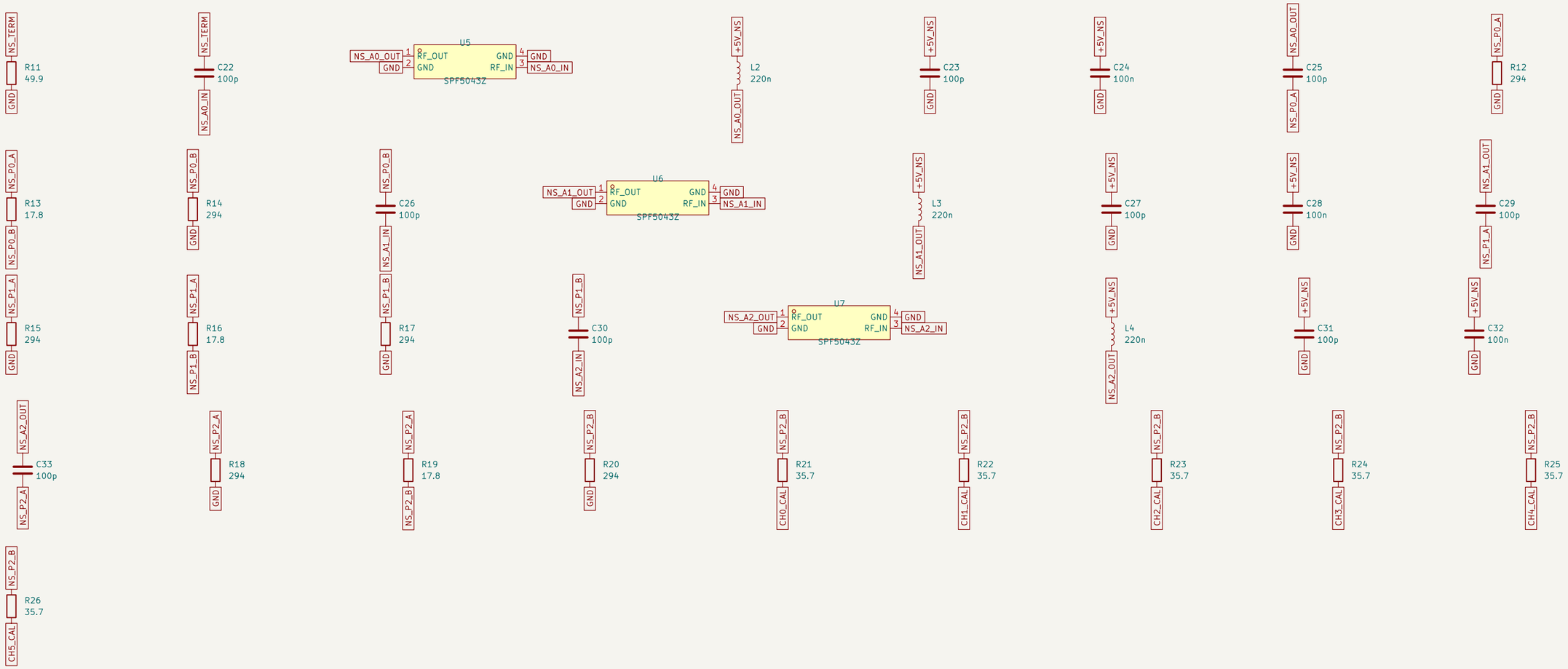
Sheet: /Power: 12 V in, 5.6 V buck, low-noise 5 V and 3.3 V/
File: power.kicad_sch

***Title: Power: 12 V in, 5.6 V buck, low-noise 5 V and 3.3 V***

| Size: User | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 2/5 |

# Calibration noise source and 6-way splitter

Expected output: about −134 dBm/Hz per channel (ENR ~39 dB) after the 15.6 dB splitter.




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**Passive radar receiver**

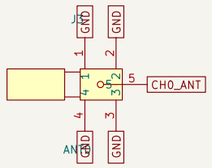
Sheet: /Calibration noise source and 6-way splitter/
File: noise_source.kicad_sch

***Title: Calibration noise source and 6-way splitter***

| Size: User | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 3/5 |

# RF channels 0–2

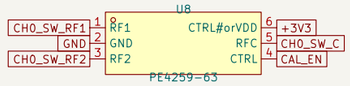
RF traces: 50 ohm grounded coplanar waveguide on L1 over solid GND on L2. Keep filter parts in line.







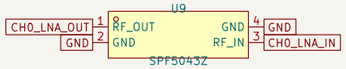




























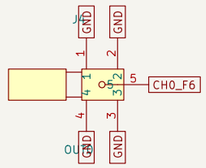


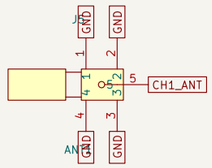








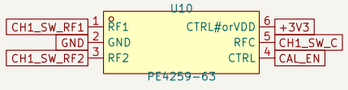





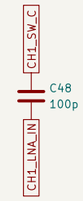


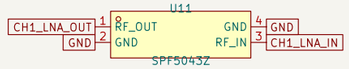
















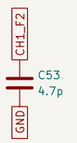












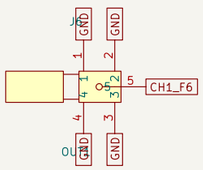


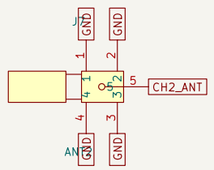








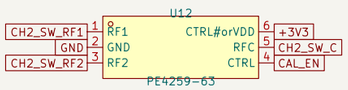

















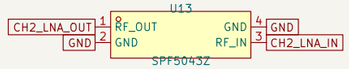












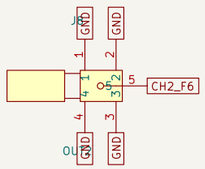














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**Passive radar receiver**

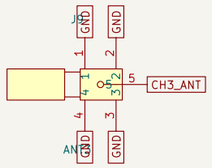
Sheet: /RF channels 0–2/
File: channels_0.kicad_sch

**Title: RF channels 0–2**

| Size: User | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 4/5 |

# RF channels 3–5

RF traces: 50 ohm grounded coplanar waveguide on L1 over solid GND on L2. Keep filter parts in line.





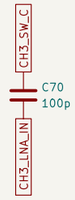


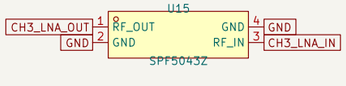




























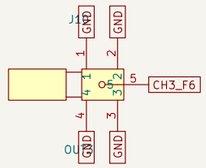


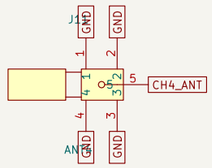








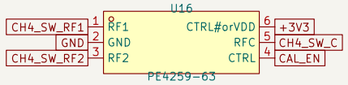



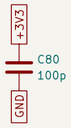



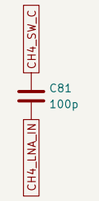


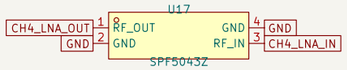




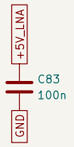























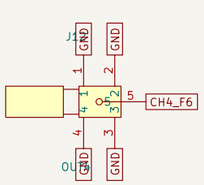


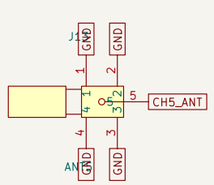








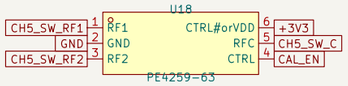



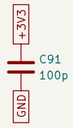







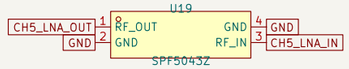




























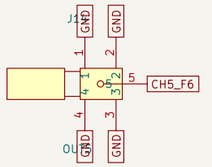











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**Passive radar receiver**

Sheet: /RF channels 3–5/
File: channels_1.kicad_sch

**Title: RF channels 3–5**

| Size: User | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 5/5 |